Manan Vijay

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Professional Summary

AI & Full-Stack Engineer specializing in LLM application development, end-to-end Retrieval-Augmented Generation (RAG) pipelines, and production backend architectures. OCI Certified Generative AI Professional with deep hands-on expertise in prompt engineering, vector embeddings (Pinecone/Chroma), semantic search optimization, and automated LLM evaluation workflows. Solved 1200+ algorithmic problems (1700+ LeetCode rating) with proven discipline in failure analysis, unit testing, Dockerized microservices, and shipping production-ready systems.

Technical Skills

AI & LLM Engineering: Retrieval-Augmented Generation (RAG), Prompt Engineering, Agent Workflows, OpenAI API, LangChain, Embeddings, Semantic Search, Failure Analysis, Test Case Evaluation, Hallucination Mitigation
Programming Languages: Python, JavaScript (ES6+), TypeScript, C++, Java, SQL, HTML5/CSS3
Vector & Relational Databases: Pinecone, ChromaDB, PostgreSQL, MongoDB, MySQL, Drizzle ORM
Backend & System Architecture: Node.js, Express.js, REST APIs, OpenAPI/Swagger, JWT Authentication, Microservices
Frontend Development: React.js, Next.js, Vite, Redux Toolkit, TanStack Query, Tailwind CSS
DevOps, Cloud & Tools: Docker, Git, GitHub Actions, CI/CD Pipelines, Postman, Linux CLI, Vercel, TDD
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, DBMS, OS, Networks

Projects

RagNite – AI RAG Platform | *Python, LangChain, OpenAI API, Pinecone, Node.js* Source/Live

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) pipeline for document ingestion, recursive text chunking, and high-dimensional vector embeddings using OpenAI APIs and Pinecone vector store.
- Developed custom prompt templates and agentic tool-calling workflows to ground LLM responses in uploaded documents, reducing hallucination rates and eliminating irrelevant context retrieval.
- Built systematic evaluation test suites to benchmark retrieval precision against complex natural language queries, turning production edge-case failures into regression test cases.
- Architected a full-stack document intelligence interface with type-safe REST APIs, JWT authentication, and Docker containerization for predictable multi-environment deployments.

InstaVEG – High-Reliability Multi-Role Commerce Platform | *TypeScript, React.js, Node.js, PostgreSQL, Drizzle ORM*

- Architected a scalable multi-role backend (Buyer, Seller, Admin) with granular RBAC; performed a self-directed security audit to identify and patch critical BOLA/IDOR authorization vulnerabilities prior to release.
- Engineered type-safe REST APIs adhering strictly to OpenAPI specifications with runtime Zod schema validation, preventing malformed payloads from reaching PostgreSQL tables.
- Shipped real-time state synchronization using TanStack Query, integrated Razorpay payment webhooks, and containerized microservices with Docker for seamless dev-to-prod deployment.

Experience

Freelance Full Stack & Software Developer Remote
Independent Engineering Client Engagements *May 2024 – Present*

- Spearheaded end-to-end development of 10+ web platforms for independent clients, owning architecture, REST API design, frontend UI, database modeling, and live cloud deployment.
- Constructed secure backend services (Node.js, Express.js) with JWT authentication, structured error-handling middleware, and comprehensive Postman integration test suites.
- Conducted proactive failure investigations, debugging production crashes and writing automated test cases to maintain 99%+ application uptime across long-term engagements.

Achievements & Certifications

- **Oracle Cloud Infrastructure (OCI) Certified Generative AI Professional** – Mastered LLM architectures, fine-tuning methodologies, RAG orchestration, and vector retrieval systems.
- **Competitive Programming Track Record**: Solved 1200+ DSA problems across platforms (800+ LeetCode, 1700+ Contest Rating); Global Rank 198 (Biweekly Contest 174) and 1576 (Biweekly Contest 175).
- **Smart India Hackathon (SIH) Finalist**: Selected nationally to engineer scalable software solutions for complex operational problem statements.
- **JPMorgan Chase & Co. Software Engineering Simulation**: Implemented backend data feeds, unit tests, and asynchronous event streams using Java and Apache Kafka.

Education

Bachelor of Technology in Electronics & Communication Engineering 2023 – 2027
JECRC Foundation (Rajasthan Technical University) | CGPA: 7.8/10 *Jaipur, Rajasthan*